

Question			Solution	Marks
6	(a)	(i)	Arrow points to the right	B1
		(ii)	(electric) field strength increases or (electric) force increases acceleration increases	B1 B1
		(iii)	force (on proton) has same the magnitude (of force on electron) force (on proton) is in opposite direction (to force on electron)	B1 B1
	(b)	(i)	Loss in KE = Gain in Electric PE $\frac{1}{2}mv^2 = qV$ $\frac{1}{2} \times 1.66 \times 10^{-27} \times v^2 = 1.60 \times 10^{-19} \times 1200$ $v = 4.8 \times 10^5 \text{ m s}^{-1}$	C1 C1 A1
		(ii)	$x = 5 \text{ cm. (4.5 cm to 5.5 cm)}$ Smallest potential gradient. Potential gradient is proportional to field strength Force is proportional to field strength	B1 B1

Question			Solution	Marks
7	(a)	(i)	$N_C / N_D = V_C / V_D$	